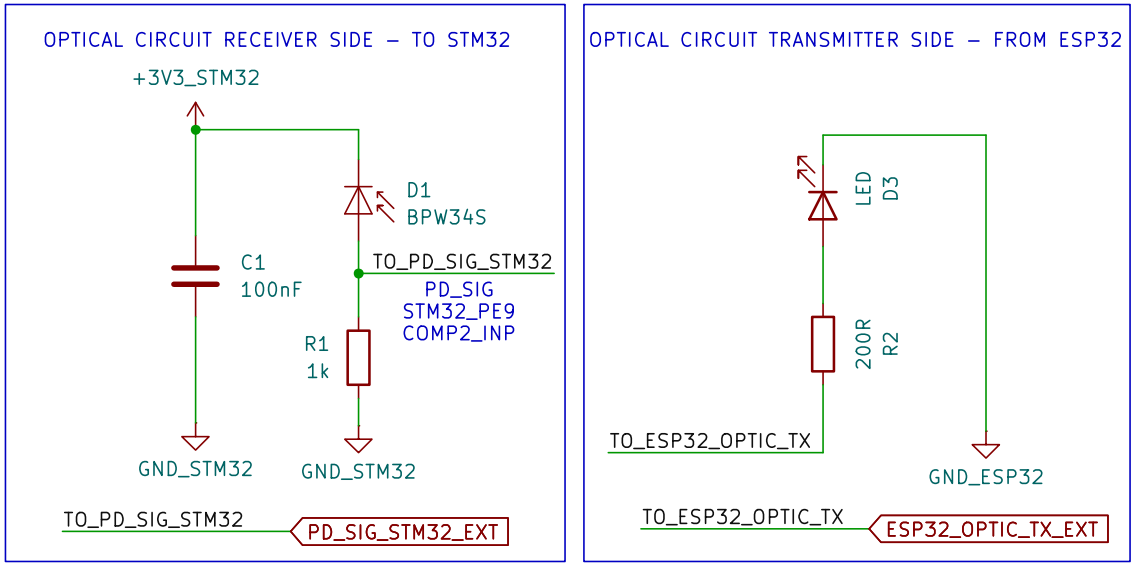
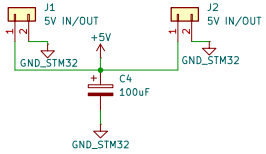
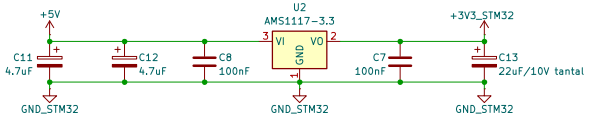
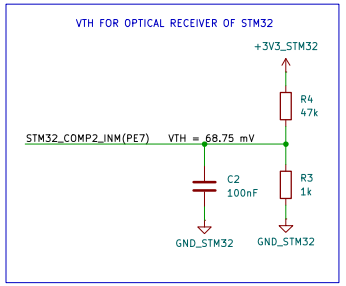
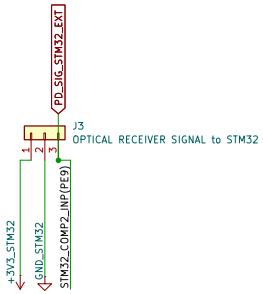

Optical transmission path
Plik: optical_path.kicad_sch
STM32 carrier board
Plik: STM32_carrier_board.kicad_sch
ESP32 carrier board
Plik: ESP32_carrier_board.kicad_sch

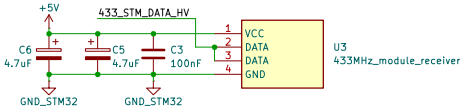

OPTICAL CIRCUIT RECEIVER SIDE – TO STM32
+3V3_STM32
D1
BPW34S
C1
100nF
TO_PD_SIG_STM32
PD_SIG
STM32_PE9
COMP2_INP
R1
1k
GND_STM32
GND_STM32
TO_PD_SIG_STM32
PD_SIG_STM32_EXT
OPTICAL CIRCUIT TRANSMITTER SIDE – FROM ESP32
LED
D3
200R
R2
TO_ESP32_OPTIC_TX
GND_ESP32
TO_ESP32_OPTIC_TX
ESP32_OPTIC_TX_EXT

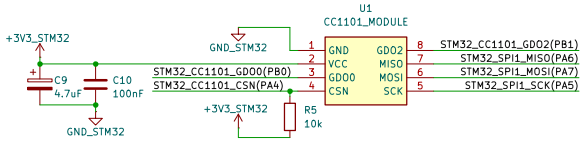

J1
5V IN/OUT
J2
5V IN/OUT
+5V
GND_STM32
GND_STM32
C4
100uF
GND_STM32

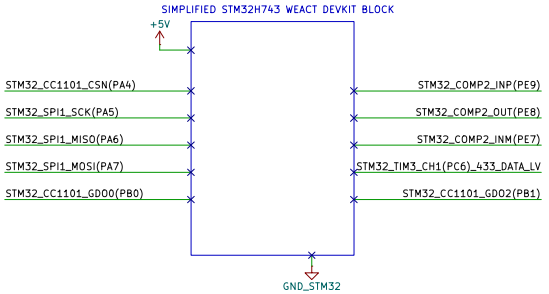

U2
AMS1117-3.3
+5V
+3V3_STM32
C11
4.7uF
C12
4.7uF
C8
100nF
C7
100nF
C13
22uF/10V tantal
VI
VO
GND
GND_STM32
GND_STM32
GND_STM32

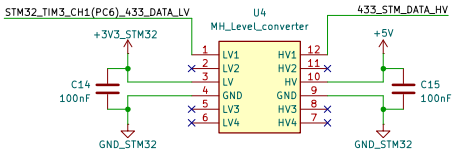

VTH FOR OPTICAL RECEIVER OF STM32
+3V3_STM32
R4
47k
STM32_COMP2_INM(PE7)
VTH = 68.75 mV
C2
100nF
R3
1k
GND_STM32
GND_STM32

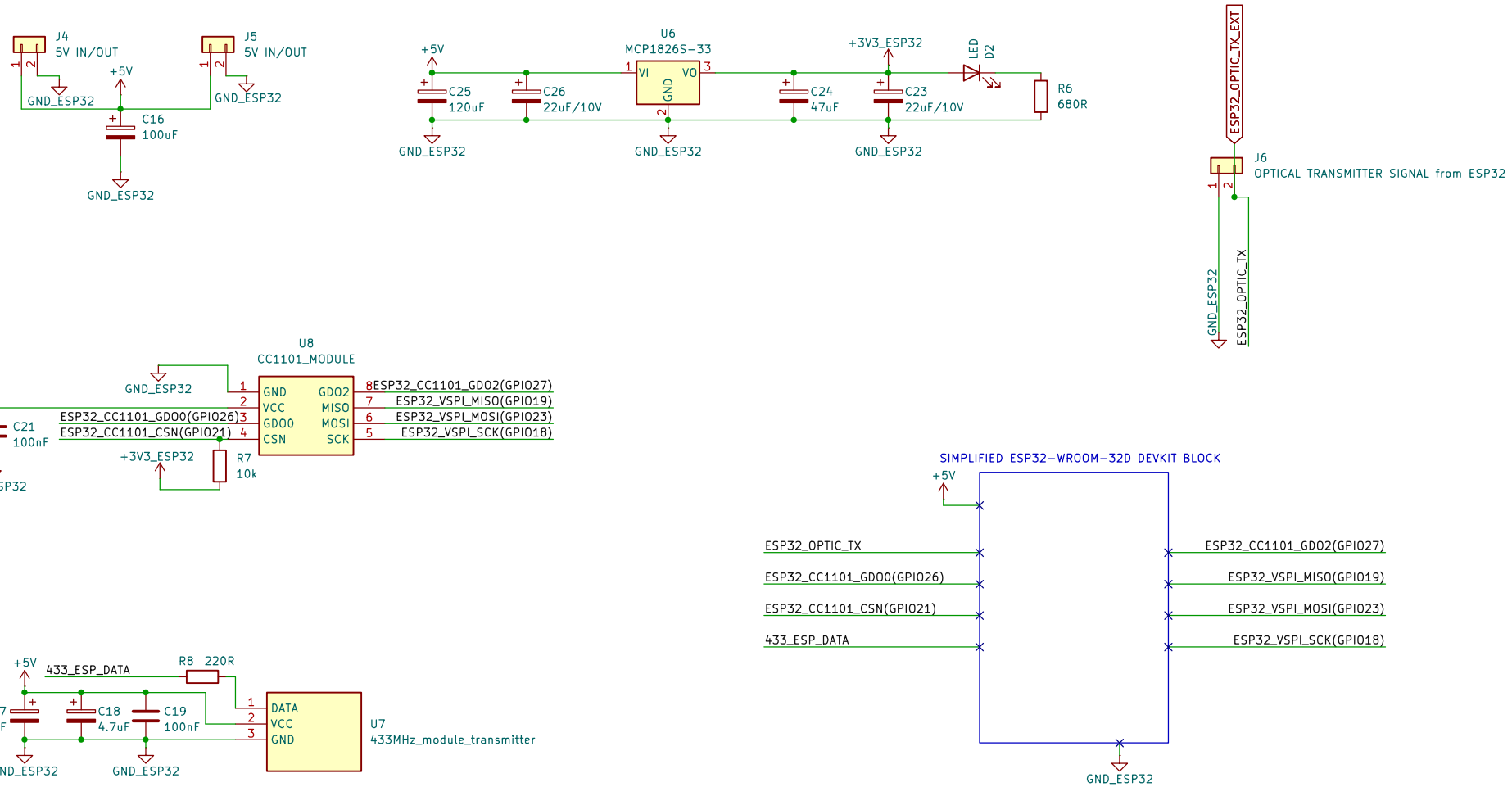

PD_SIG_STM32_EXT
J3
OPTICAL RECEIVER SIGNAL to STM32
+3V3_STM32
GND_STM32
STM32_COMP2_INP(PE9)


+5V
433_STM_DATA_HV
C6
4.7uF
C5
4.7uF
C3
100nF
VCC
DATA
DATA
GND
U3
433MHz_module_receiver
GND_STM32
GND_STM32


U1
CC1101_MODULE
+3V3_STM32
GND_STM32
C9
4.7uF
C10
100nF
STM32_CC1101_GDO0(PB0)
STM32_CC1101_CSN(PA4)
GND
VCC
GDO0
CSN
GDO2
MISO
MOSI
SCK
STM32_CC1101_GDO2(PB1)
STM32_SPI1_MISO(PA6)
STM32_SPI1_MOSI(PA7)
STM32_SPI1_SCK(PA5)
+3V3_STM32
R5
10k
GND_STM32


SIMPLIFIED STM32H743 WEACT DEVKIT BLOCK
+5V
STM32_CC1101_CSN(PA4)
STM32_SPI1_SCK(PA5)
STM32_SPI1_MISO(PA6)
STM32_SPI1_MOSI(PA7)
STM32_CC1101_GDO0(PB0)
STM32_COMP2_INP(PE9)
STM32_COMP2_OUT(PE8)
STM32_COMP2_INM(PE7)
STM32_TIM3_CH1(PC6)_433_DATA_LV
STM32_CC1101_GDO2(PB1)
GND_STM32


STM32_TIM3_CH1(PC6)_433_DATA_LV
U4
MH_Level_converter
433_STM_DATA_HV
+3V3_STM32
+5V
C14
100nF
C15
100nF
LV1
LV2
LV
GND
LV3
LV4
HV1
HV2
HV
GND
HV3
HV4
GND_STM32
GND_STM32


J4
5V IN/OUT
J5
5V IN/OUT
+5V
GND_ESP32
C16
100uF
U6
MCP1826S-33
VI
VO
GND
C25
120uF
C26
22uF/10V
C24
47uF
C23
22uF/10V
+3V3_ESP32
LED
D2
R6
680R
ESP32_OPTIC_TX_EXT
J6
OPTICAL TRANSMITTER SIGNAL from ESP32
ESP32_OPTIC_TX
U8
CC1101_MODULE
+3V3_ESP32
C20
4.7uF
C21
100nF
GND
VCC
GDO0
CSN
GDO2
MISO
MOSI
SCK
ESP32_CC1101_GDO0(GPIO26)
ESP32_CC1101_CSN(GPIO21)
ESP32_CC1101_GDO2(GPIO27)
ESP32_VSPI_MISO(GPIO19)
ESP32_VSPI_MOSI(GPIO23)
ESP32_VSPI_SCK(GPIO18)
R7
10k
SIMPLIFIED ESP32-WROOM-32D DEVKIT BLOCK
ESP32_OPTIC_TX
433_ESP_DATA
R8 220R
C17
4.7uF
C18
4.7uF
C19
100nF
DATA
U7
433MHz_module_transmitter